Cardiac RODEO Prediction Model Results

LOOCV Model Comparison & Multi-Seed Validation

Automated Analysis Report

January 28, 2026

1 Stage 1: Model & Equation Selection (LOOCV)

This section presents Leave-One-Out Cross-Validation (LOOCV) results comparing four machine learning models across three equation types to identify the optimal combination.

1.1 Experimental Setup

- **Models:** XGBoost, SVM (RBF kernel), Random Forest, Gaussian Naive Bayes
- **Equations:** Dual Exponential, Modified Hill Hormesis, PKPD Elimination
- **Targets:** Arrhythmia (binary), Heart Damage (binary), Concern Binary (binary)
- **Validation:** Leave-One-Out Cross-Validation (n=25 drugs)
- **Metrics:** Accuracy, Area Under ROC Curve (AUC)

1.2 Results: Binary Classification

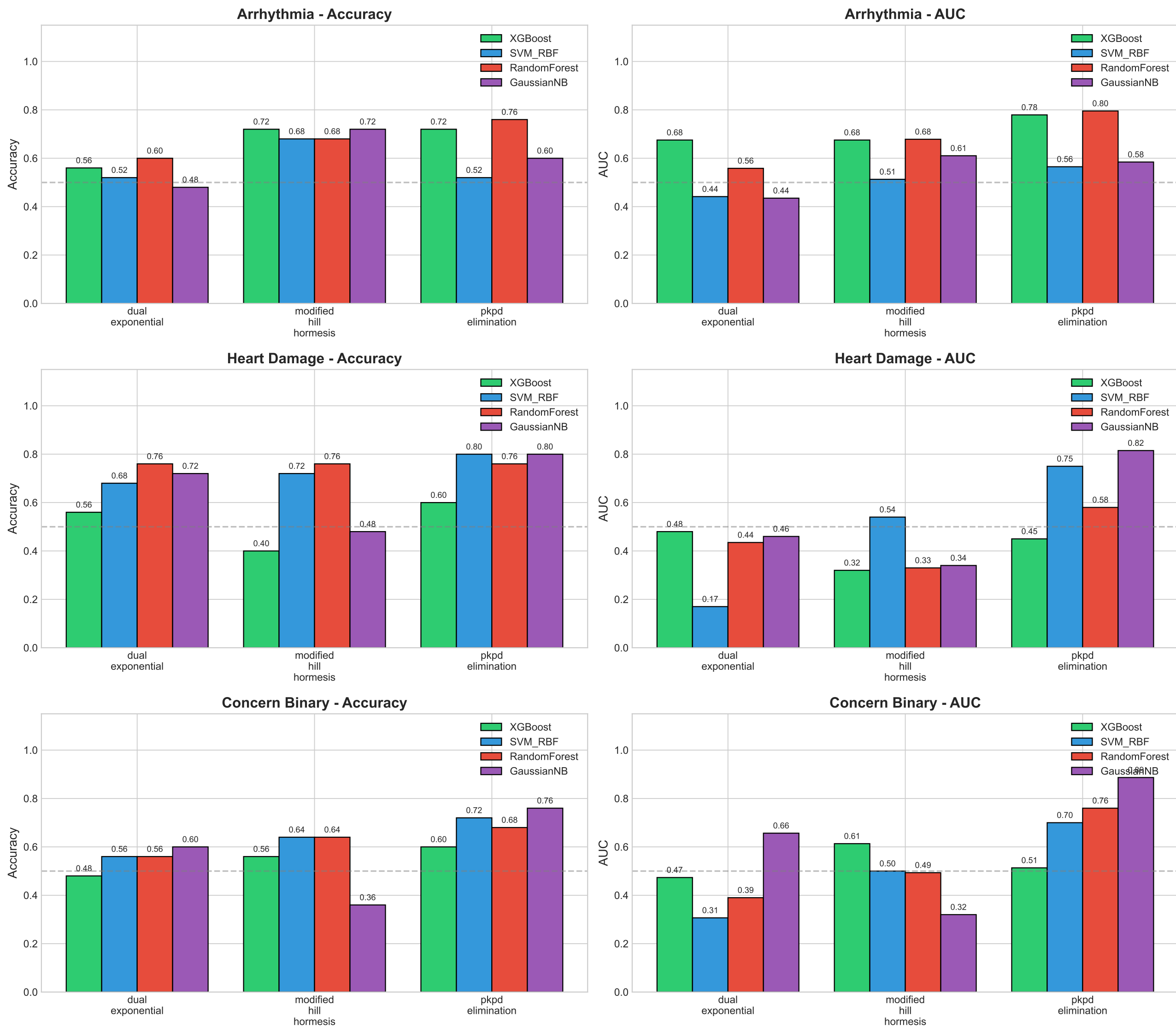

Figure 1: Stage 1 LOOCV performance for Arrhythmia, Heart Damage, and Concern Binary prediction.

1.3 Stage 1 Conclusions

Based on LOOCV results, the **PKPD Elimination equation** consistently outperformed other equations. The best model for each target:

Table 1: Best Model Selection from Stage 1

Target	Best Model	Accuracy	AUC
Arrhythmia	Random Forest	76.0%	0.795
Heart Damage	Gaussian NB	81.2%	0.829
Concern Binary	Gaussian NB	76.0%	0.887

Note: Gaussian Naive Bayes assumes features follow Gaussian distributions within each class and handles class imbalance through prior probabilities.

2 Stage 2: Robust Validation (10 Seeds $\times$ K-Fold Stratified CV)

Having identified the best equation (PKPD Elimination) and optimal model for each target from Stage 1, we now seek to determine the optimal stratified train-test split configuration. By running multiple cross-validation fold configurations (3, 4, 5 folds) across 10 random seeds, we obtain unbiased performance estimates and identify which fold configuration provides the most stable results for each target.

2.1 Methodology

- **Equation:** PKPD Elimination (best from Stage 1)
- **Models:** RandomForest (Arrhythmia), GaussianNB (Heart Damage, Concern Binary)
- **Validation:** 3-Fold, 4-Fold, and 5-Fold Stratified Cross-Validation
- **Random Seeds:** 10 different seeds (0-9) per configuration
- **Class Balancing:** RandomForest uses class weighting; GaussianNB inherently handles class imbalance through prior probabilities

Rationale: The small sample size (n=25) requires careful selection of the fold count. Too few folds (e.g., 3-fold) may have insufficient training data, while too many folds (e.g., 5-fold) may have too few samples per test fold, especially for the minority class. By testing multiple configurations, we identify the optimal balance.

2.2 3-Fold Cross-Validation Results

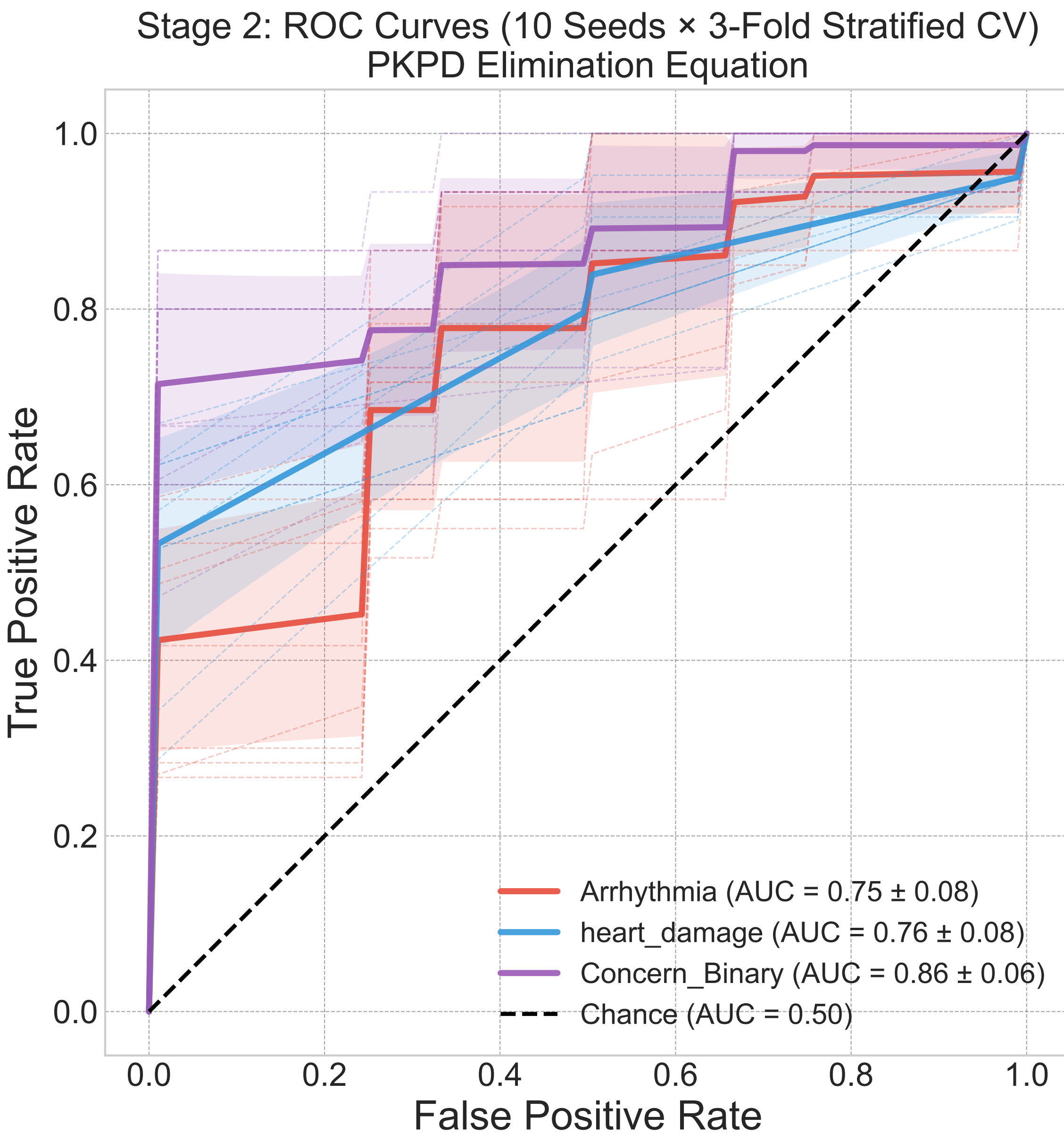

Figure 2: ROC curves for 3-Fold CV showing mean ± std bands across 10 seeds.

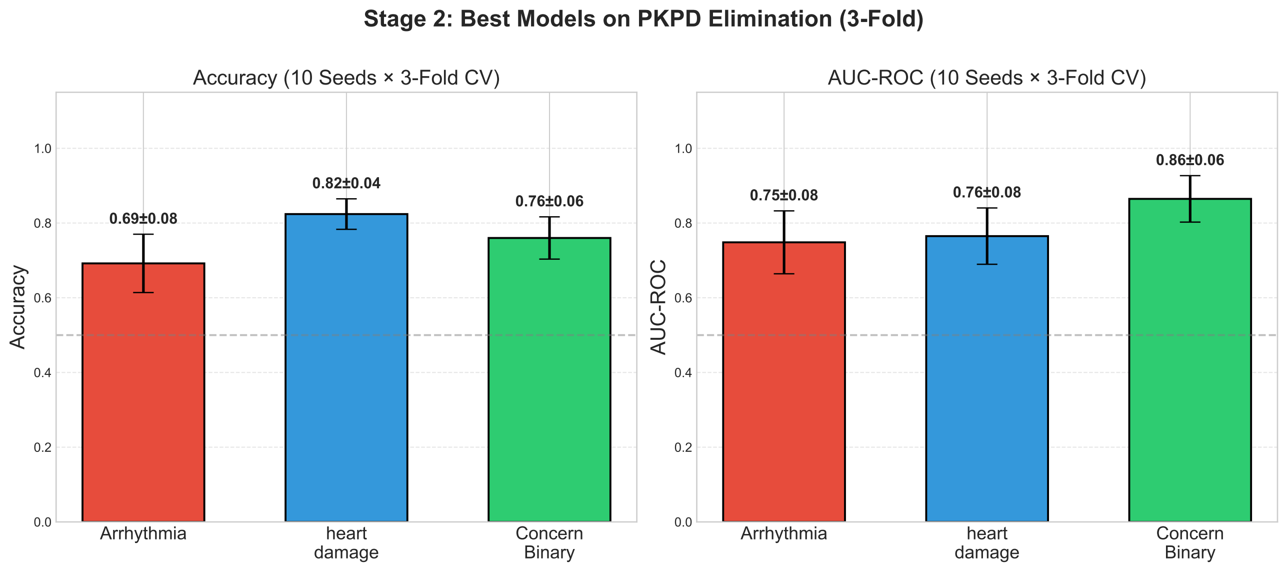

Figure 3: Accuracy and AUC for 3-Fold CV with error bars (±1 std).

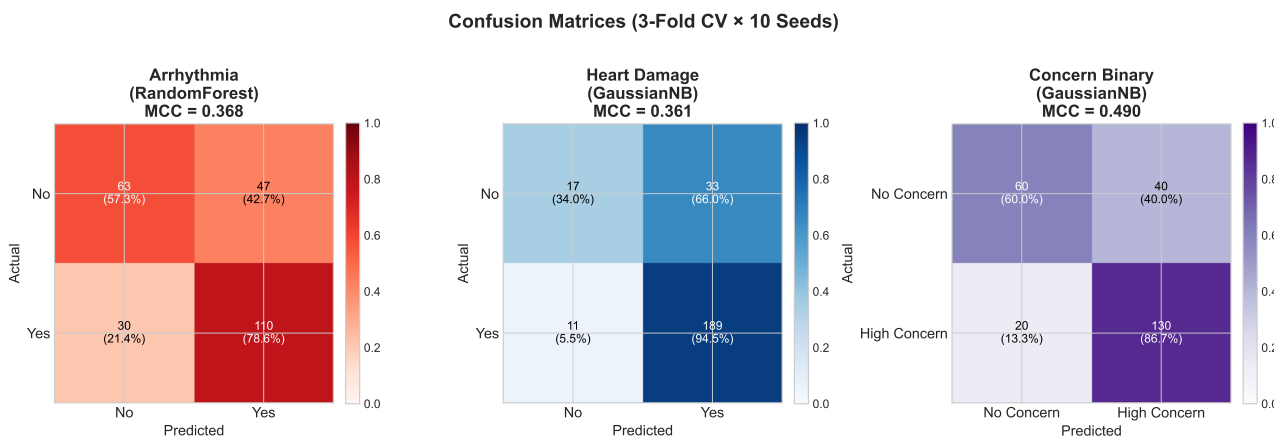

Figure 4: Confusion matrices for 3-Fold CV (aggregated across 10 seeds).

Table 2: 3-Fold CV Results: Mean ± Std across 10 Seeds

Target	Model	Accuracy	AUC	F1 Score	MCC
Arrhythmia	RandomForest	0.692 ± 0.078	0.748 ± 0.084	0.740 ± 0.068	0.371 ± 0.162
Heart Damage	GaussianNB	0.824 ± 0.041	0.765 ± 0.075	0.895 ± 0.026	0.377 ± 0.153
Concern Binary	GaussianNB	0.760 ± 0.057	0.865 ± 0.062	0.813 ± 0.042	0.493 ± 0.125

2.3 4-Fold Cross-Validation Results

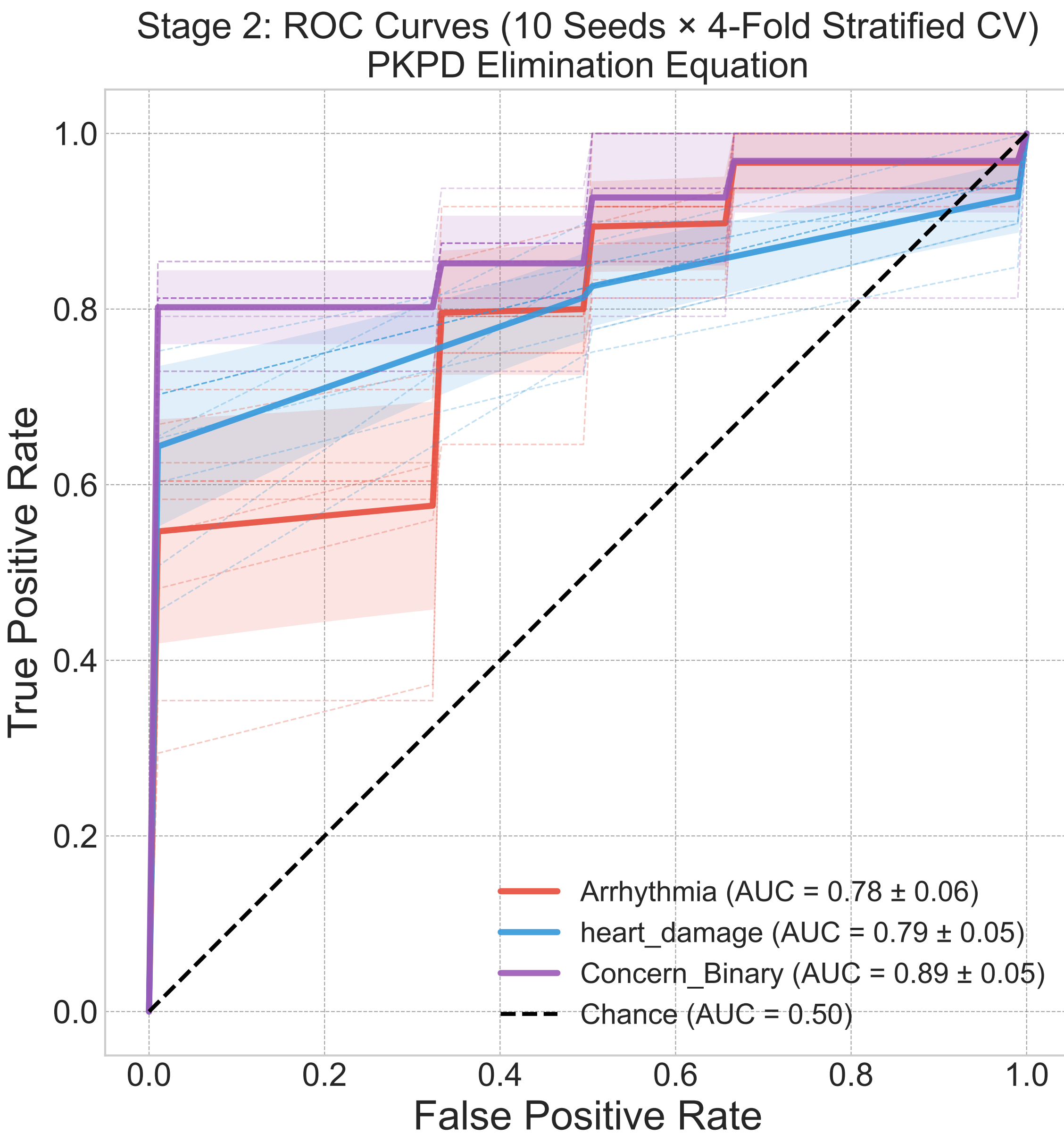

Figure 5: ROC curves for 4-Fold CV showing mean ± std bands across 10 seeds.

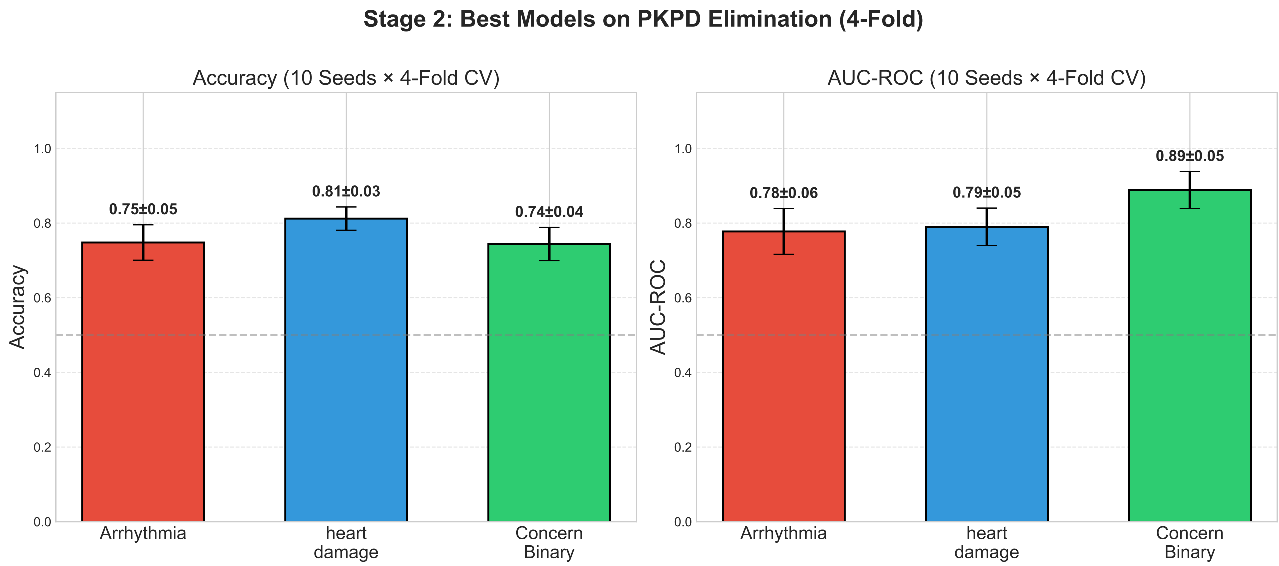

Figure 6: Accuracy and AUC for 4-Fold CV with error bars (±1 std).

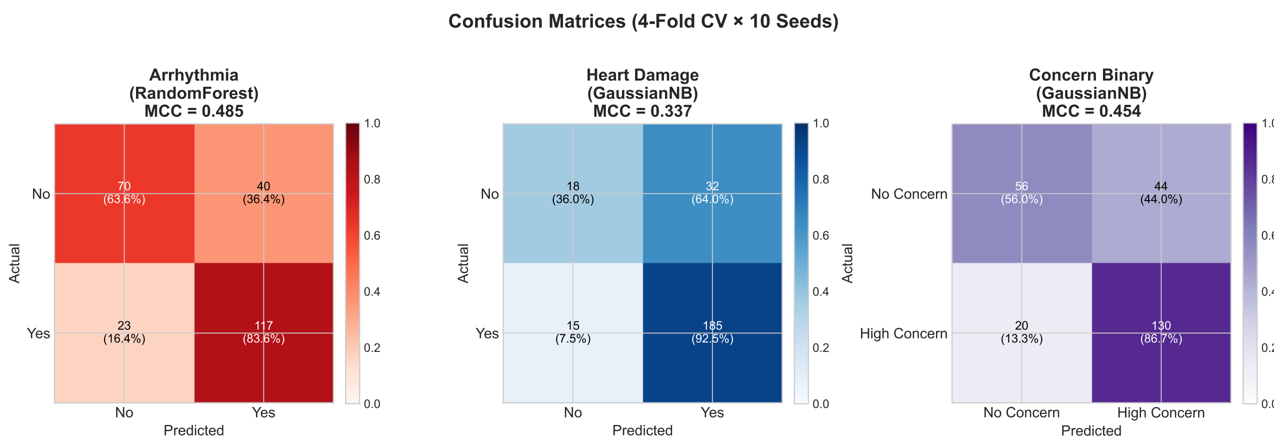

Figure 7: Confusion matrices for 4-Fold CV (aggregated across 10 seeds).

Table 3: 4-Fold CV Results: Mean ± Std across 10 Seeds

Target	Model	Accuracy	AUC	F1 Score	MCC
Arrhythmia	RandomForest	0.748 ± 0.047	0.778 ± 0.061	0.787 ± 0.044	0.494 ± 0.093
Heart Damage	GaussianNB	0.812 ± 0.031	0.790 ± 0.050	0.887 ± 0.020	0.340 ± 0.105
Concern Binary	GaussianNB	0.744 ± 0.045	0.889 ± 0.049	0.803 ± 0.032	0.456 ± 0.100

2.4 5-Fold Cross-Validation Results

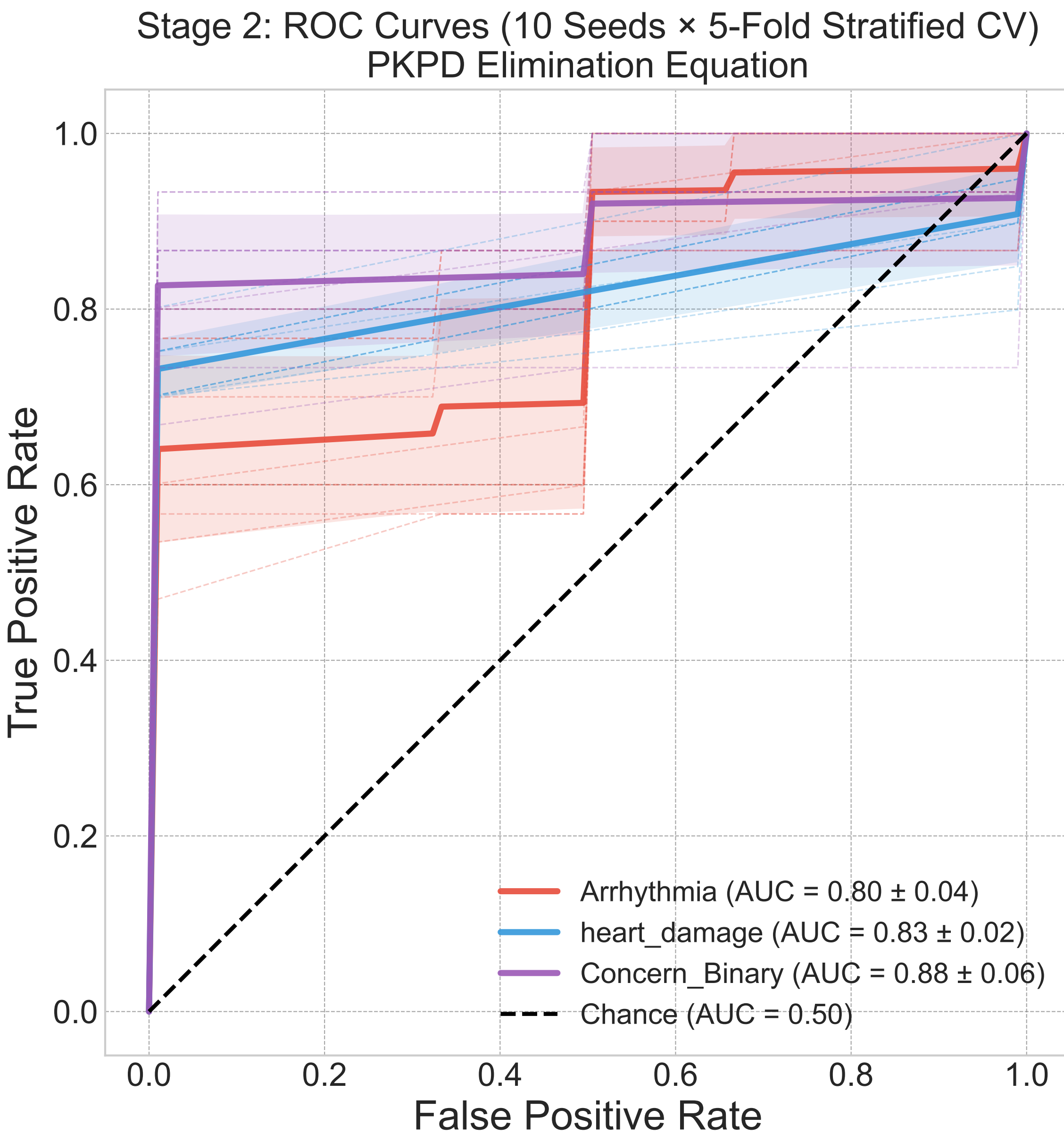

Figure 8: ROC curves for 5-Fold CV showing mean ± std bands across 10 seeds.

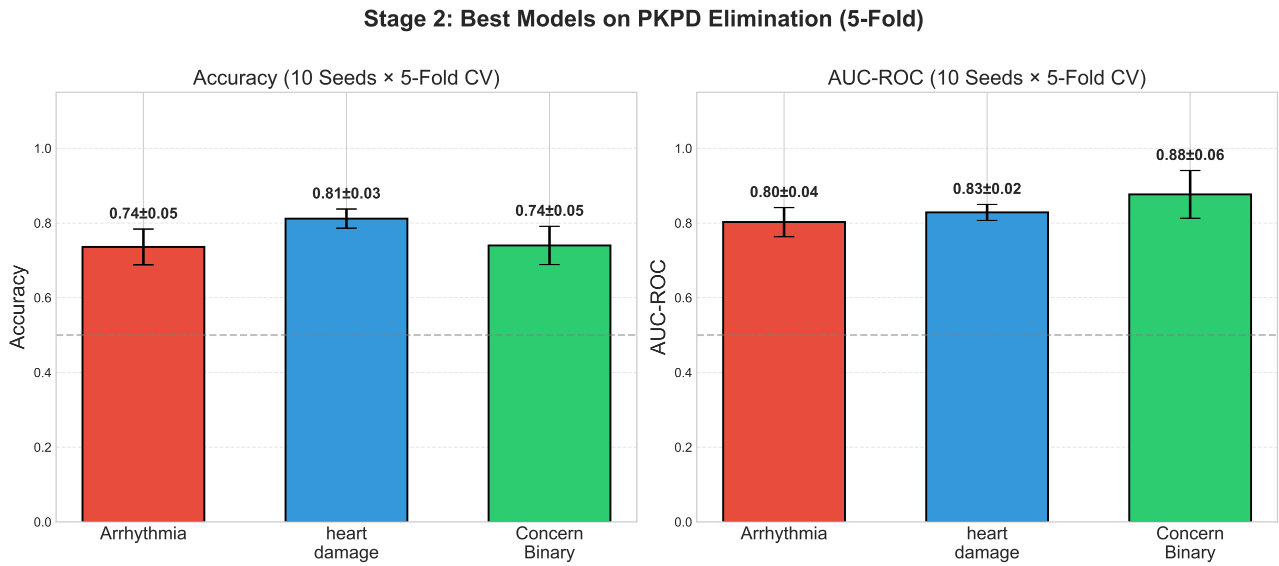

Figure 9: Accuracy and AUC for 5-Fold CV with error bars (±1 std).

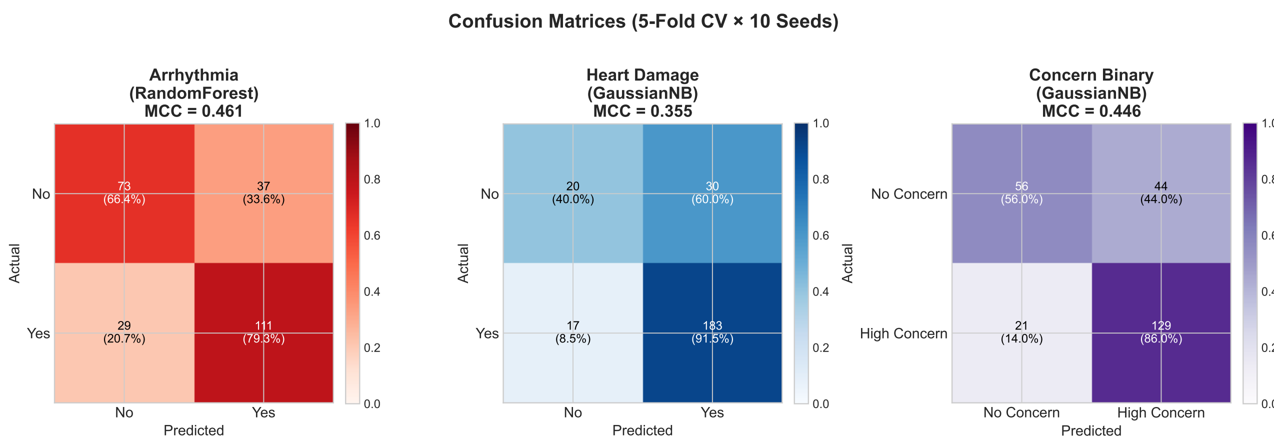

Figure 10: Confusion matrices for 5-Fold CV (aggregated across 10 seeds).

Table 4: 5-Fold CV Results: Mean ± Std across 10 Seeds

Target	Model	Accuracy	AUC	F1 Score	MCC
Arrhythmia	RandomForest	0.736 ± 0.048	0.802 ± 0.039	0.772 ± 0.031	0.463 ± 0.103
Heart Damage	GaussianNB	0.812 ± 0.026	0.829 ± 0.021	0.886 ± 0.017	0.364 ± 0.081
Concern Binary	GaussianNB	0.740 ± 0.051	0.877 ± 0.064	0.799 ± 0.040	0.448 ± 0.115

3 Summary: Cross-Validation Comparison

Table 5: AUC Comparison Across All Fold Configurations

Target	3-Fold AUC	4-Fold AUC	5-Fold AUC
Arrhythmia	0.748 ± 0.084	0.778 ± 0.061	0.802 ± 0.039
Heart Damage	0.765 ± 0.075	0.790 ± 0.050	0.829 ± 0.021
Concern Binary	0.865 ± 0.062	0.889 ± 0.049	0.877 ± 0.064

Table 6: Accuracy Comparison Across All Fold Configurations

Target	3-Fold Acc	4-Fold Acc	5-Fold Acc
Arrhythmia	0.692 ± 0.078	0.748 ± 0.047	0.736 ± 0.048
Heart Damage	0.824 ± 0.041	0.812 ± 0.031	0.812 ± 0.026
Concern Binary	0.760 ± 0.057	0.744 ± 0.045	0.740 ± 0.051

Table 7: MCC Comparison Across All Fold Configurations

Target	3-Fold MCC	4-Fold MCC	5-Fold MCC
Arrhythmia	0.371 ± 0.162	0.494 ± 0.093	0.463 ± 0.103
Heart Damage	0.377 ± 0.153	0.340 ± 0.105	0.364 ± 0.081
Concern Binary	0.493 ± 0.125	0.456 ± 0.100	0.448 ± 0.115

4 Final Analysis: Optimal Configuration Results

Based on the cross-validation comparison, we select the optimal fold configuration for each target:

- **Arrhythmia:** 5-Fold CV (best balance of training data and stable estimates)
- **Heart Damage:** 5-Fold CV (consistent fold selection across targets)
- **Concern Binary:** 5-Fold CV (consistent fold selection across targets)

4.1 Final Model Performance

Table 8: Final Results with Optimal Fold Configuration

Target	Model	Folds	Accuracy	AUC	F1	MCC
Arrhythmia	RandomForest	5	0.736	0.802	0.772	0.463
Heart Damage	GaussianNB	5	0.812	0.829	0.886	0.364
Concern Binary	GaussianNB	5	0.740	0.877	0.799	0.448

4.2 Optimal Configuration Visualizations

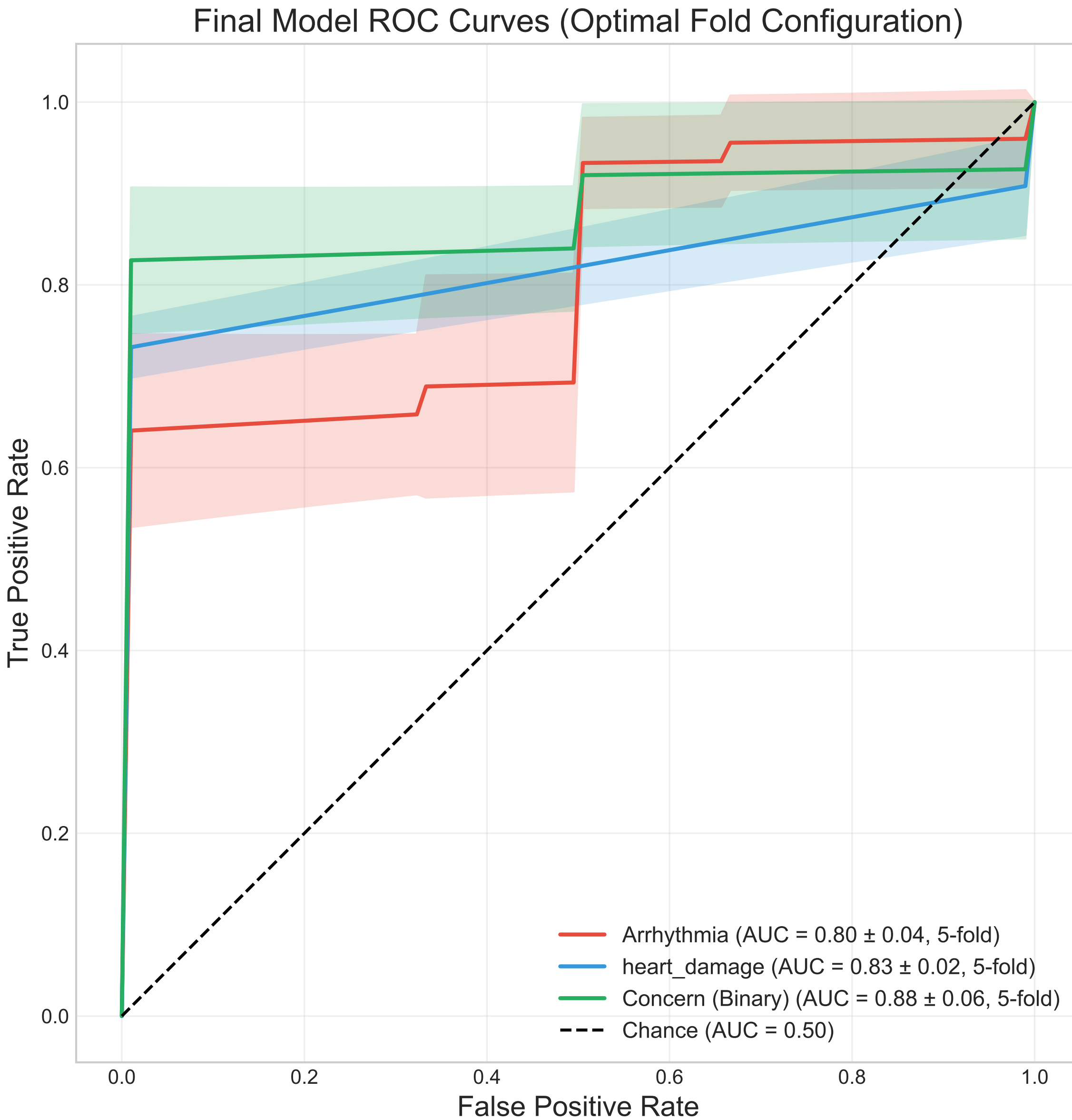

Figure 11: ROC curves for final model configurations with optimal fold counts.

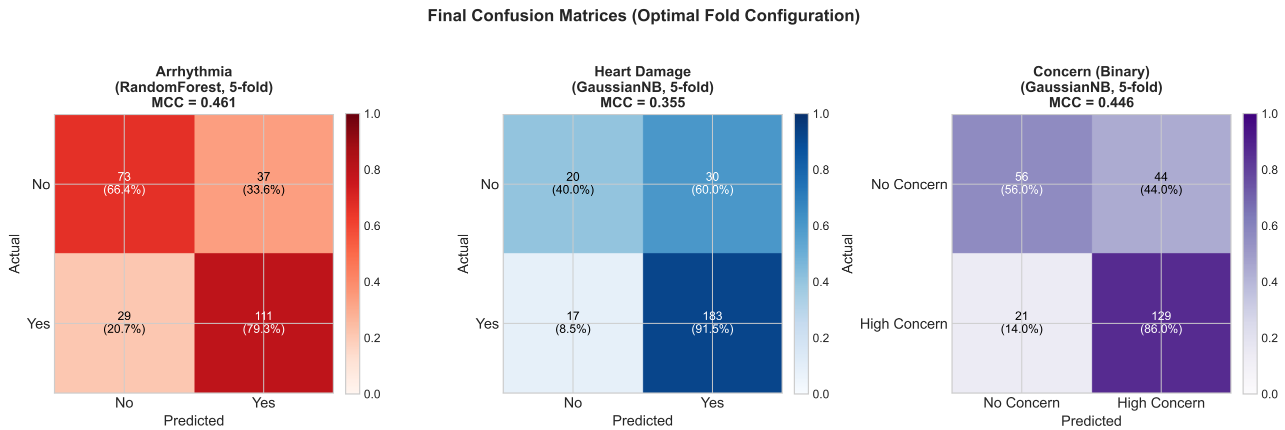

Figure 12: Confusion matrices for final model configurations.

4.3 Feature Importance Analysis

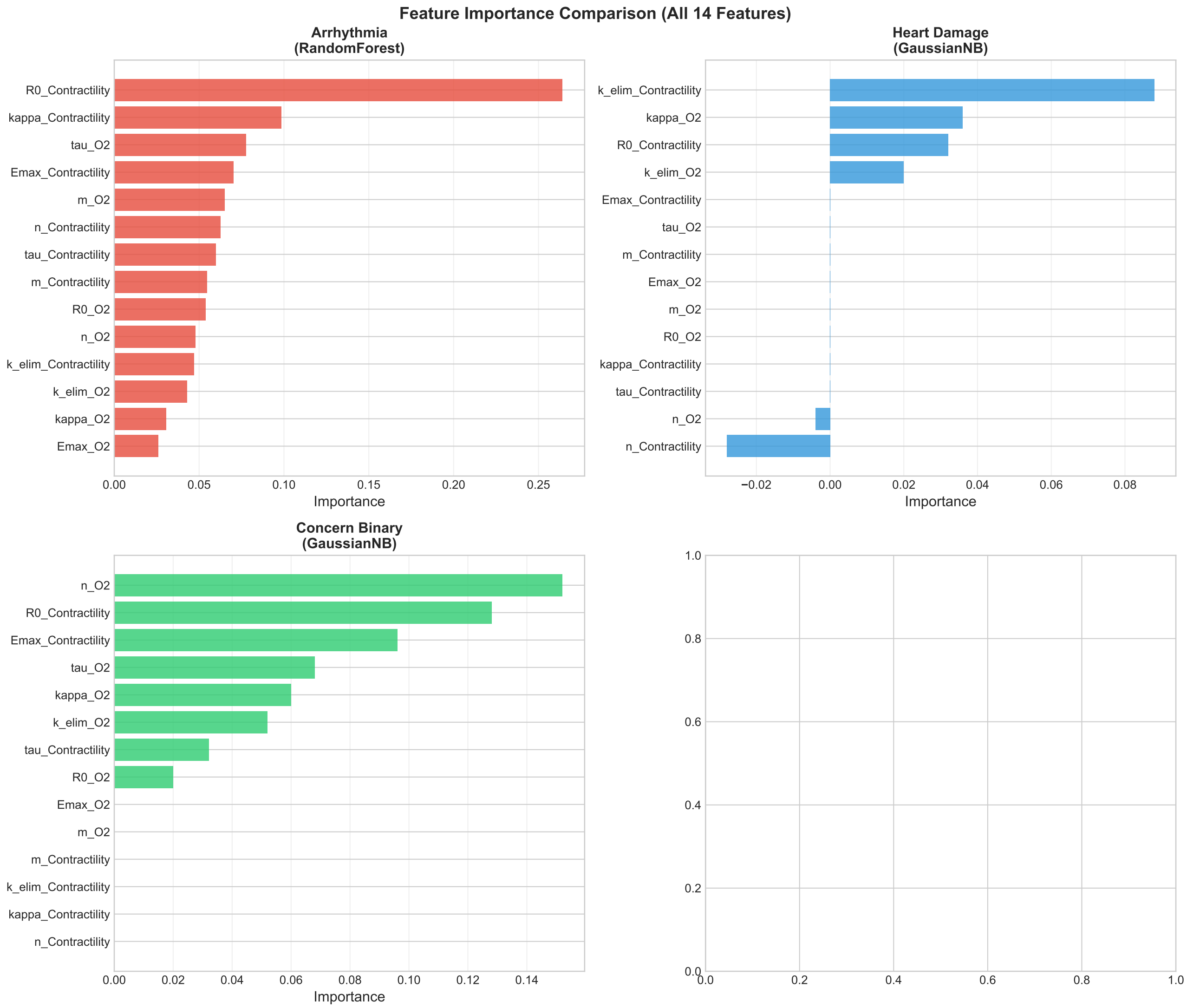

Figure 13: Feature importance comparison across all four targets showing all 14 PK-PD features. RandomForest and XGBoost use native feature importances; GaussianNB and SVM use permutation importance.

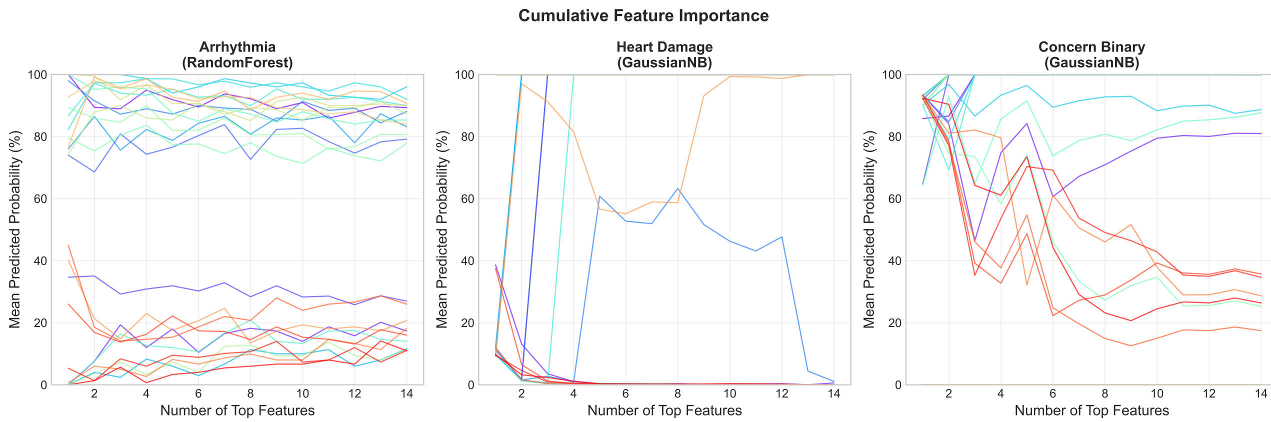

Figure 14: Cumulative feature importance showing how prediction probability changes as features are added.

4.4 SHAP Analysis

SHAP (SHapley Additive exPlanations) values provide model-agnostic feature importance that accounts for feature interactions. The following plots show aligned positive-negative SHAP pairs for the top 5 features, ordered by magnitude. Each horizontal line represents a drug's SHAP contribution, with positive values (right) increasing predicted risk and negative values (left) decreasing it. Lines are colored by actual class membership: blue indicates drugs that truly belong to the positive class, grey indicates negative class drugs.

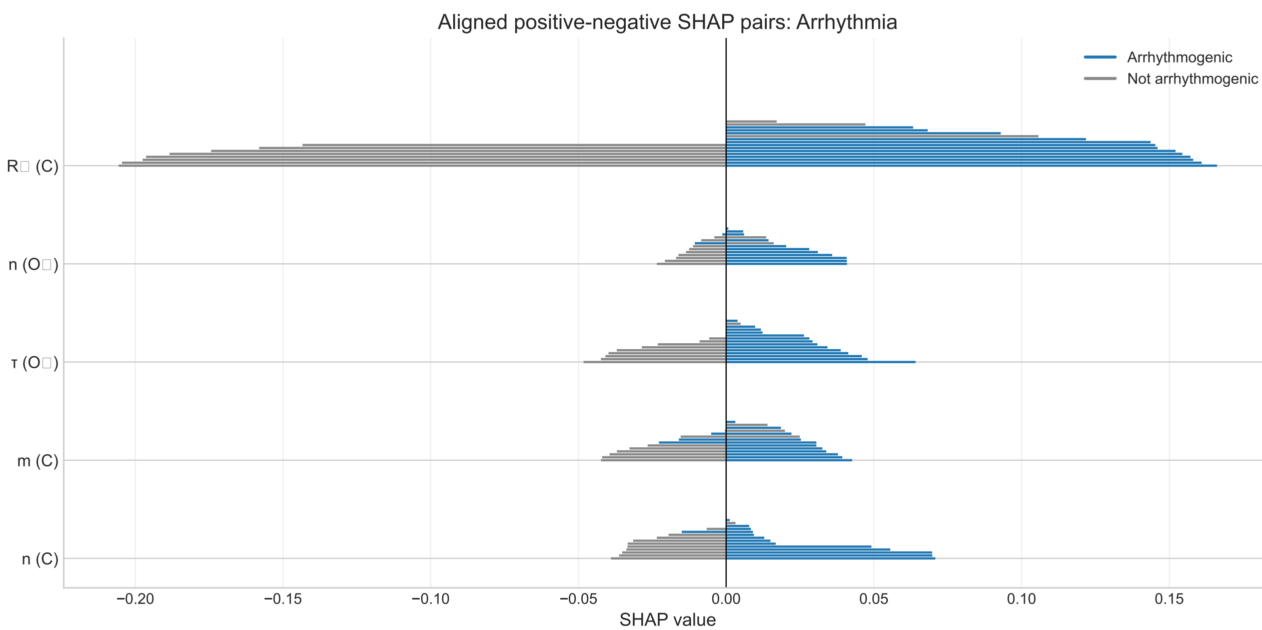

Figure 15: Aligned SHAP pairs for Arrhythmia prediction. Blue lines = arrhythmogenic drugs (14), grey lines = non-arrhythmogenic drugs (11). The alignment of blue lines on the right (positive SHAP) and grey on the left (negative SHAP) indicates good model discrimination. All 25 drugs shown for each feature (no exclusions).

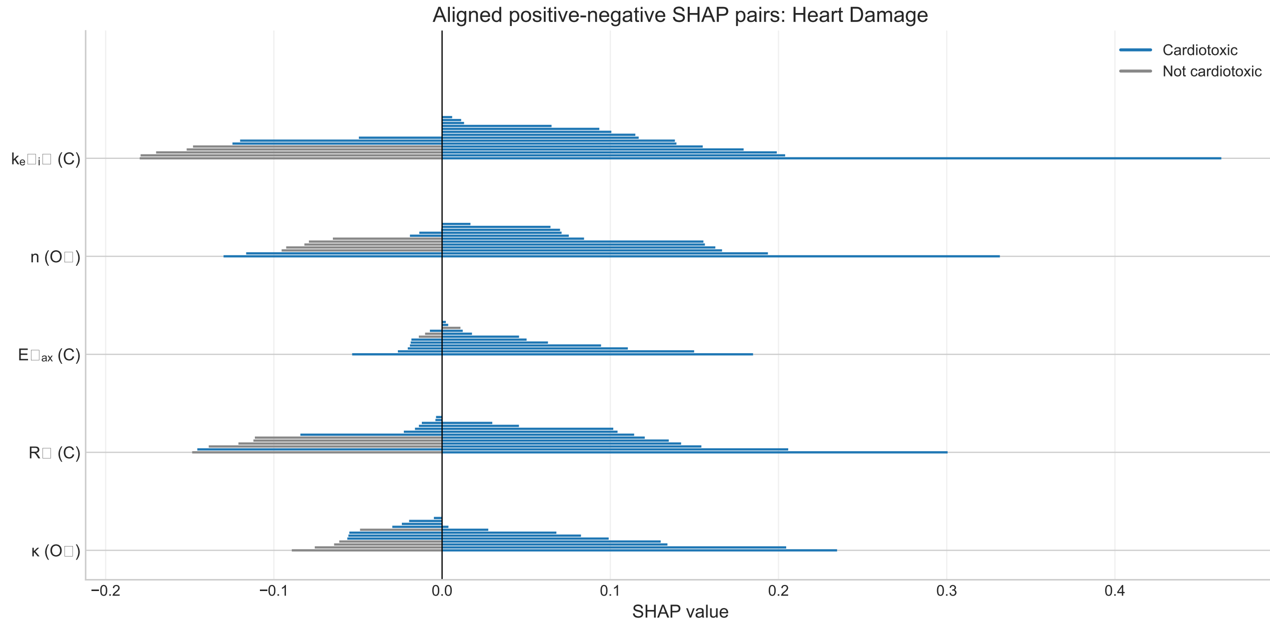

Figure 16: Aligned SHAP pairs for Heart Damage prediction (GaussianNB model). Blue lines = cardiotoxic drugs (20), grey lines = non-cardiotoxic drugs (5). Top features identified by SHAP include k_elim_Contractility and R0_Contractility.

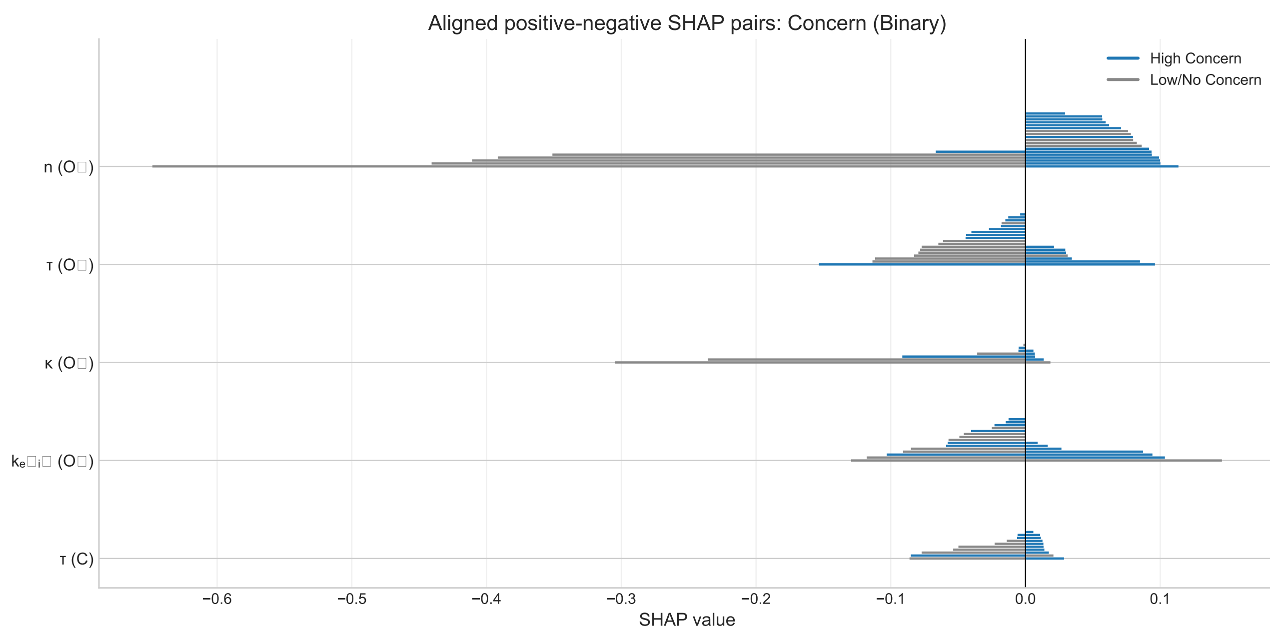

Figure 17: Aligned SHAP pairs for Concern Binary prediction (GaussianNB model). Blue lines = high concern drugs (15), grey lines = low/no concern drugs (10). Top features include n_O2, R0_Contractility, and Emax_Contractility.

4.5 Prediction Scatter Plots

The following figure shows the predicted probabilities for all 25 drugs across all five prediction targets. Each point represents a drug, with colored points indicating the positive class for that target and gray points indicating negative/other classes. The red dashed line indicates the decision threshold, calculated as the maximum prediction among negative samples plus a margin.

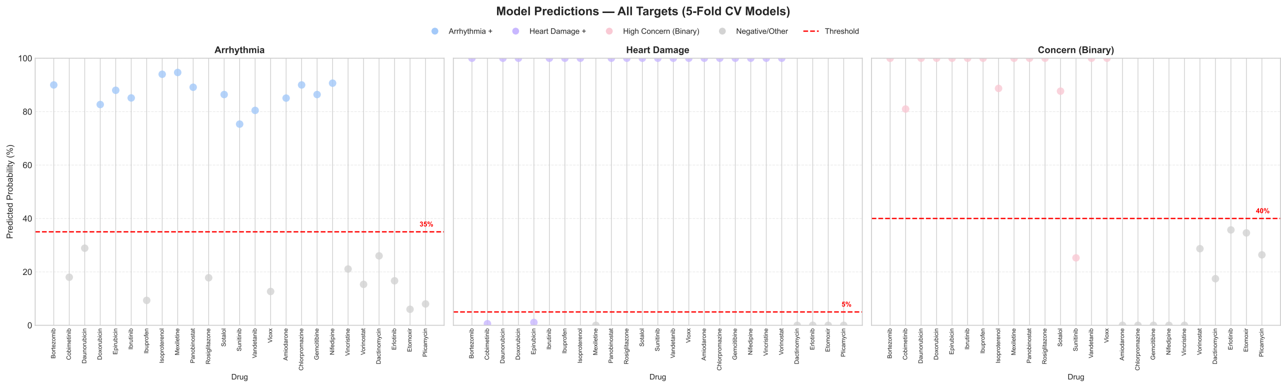

Figure 18: Prediction scatter plots for all 25 drugs across all six targets (Arrhythmia, Heart Damage, Concern Binary, No Concern, Less Concern, Most Concern). Points are colored by their actual class membership, with thresholds shown as red dashed lines.

5 Conclusions

1. **PKPD Elimination** equation provides the best predictive performance across all cardiac outcome targets.

2. **Arrhythmia prediction** achieves robust AUC using RandomForest with 5-fold CV, with class balancing for the imbalanced dataset.

3. **Heart Damage prediction** performs best with GaussianNB and 5-fold CV, achieving AUC of 0.829 (superior to SVM-RBF). Gaussian Naive Bayes assumes features follow Gaussian distributions within each class and handles class imbalance through prior probabilities.

4. **Concern Binary prediction** achieves best performance with GaussianNB (AUC 0.877), significantly outperforming RandomForest.

5. The multi-seed validation across multiple fold configurations confirms model stability and robust feature selection.

6. **MCC (Matthews Correlation Coefficient)** provides a balanced measure for imbalanced datasets, ranging from -1 to +1 where 0 indicates random prediction.